# Supporting information

Structural limits of single-barrier reform in algorithmic recourse: a formal series-system model with implications for digital health

A.C. Demidont

## Contents

## S1 Note. Scope of claims

Throughout, numerical results (baseline 0.0018%, maximum single-barrier gain 0.0054%, three-way share 87.6%, and all Shapley/Sobol/Morris/bootstrap outputs) are **level-2** model illustrations generated from the chosen topology and parameterization; they are consistent with, but not independent confirmation of, Propositions 1–2 (level 1). Pass probabilities (Table S-Deriv) are **level-3** provisional calibration assumptions. Statements about real recourse behavior or policy are **level-4** hypotheses.

## S-Deriv Table. Parameter derivation audit trail (all 11 barriers)

**Table 1. Parameter derivation audit trail.** For each barrier: target construct, source domain, exact source statistic, mapping rule, resulting pass probability, plausibility range, and transportability limitation. Auditable from this package alone; mirrored in `data/parameter_sources/parameter_derivation.md`. Source page/table references marked [verify] are to be confirmed at finalization.

| Barrier (layer) | Source domain | Exact statistic [source] | $p$ | Range | Mapping rule | Transportability limitation |
|---|---|---|---|---|---|---|
| Rapid Data Transmission (L1) | Credit reporting | Furnishers transmit adverse data to CRAs within 1–2 days [CFPB 2022, verify] | 0.30 | 0.20–0.40 | ∼30% intervene before propagation | Credit-furnishing timing; not measured for clinical/EHR data flows |
| Multi-System Integration (L1) | Credit reporting | 20% had error on ≥1 of 3 reports [FTC 2013, verify] | 0.55 | 0.40–0.65 | Complement of cross-system error rate | Cross-bureau error; not clinical multi-system integration |
| Permanent Storage (L1) | Credit reporting | 7-yr retention; <20% adverse tradelines removed in 4 yr [CFPB 2022; FCRA §605, verify] | 0.45 | 0.35–0.55 | ∼45% encounter unexpired adverse data | FCRA retention regime; not healthcare record retention |
| Error Detection Difficulty (L2) | Credit reporting | 26% identified material errors [FTC 2013, verify] | 0.35 | 0.25–0.45 | ∼35% identify specific errors | Consumer credit review; not clinical error detection |
| Correction Process Barriers (L2) | Credit reporting | 37% of disputes fully resolved [FTC 2013; CFPB 2022, verify] | 0.35 | 0.25–0.45 | ∼35% achieve full correction | Credit dispute process; not clinical data correction |
| Incomplete Correction Propagation (L2) | Credit reporting | Corrections do not auto-propagate across CRAs [CFPB 2022; FTC 2015, verify] | 0.40 | 0.30–0.50 | ∼40% propagate sufficiently | Inter-CRA propagation; not clinical systems |
| Awareness Gap (L3) | Civil legal | Help sought for ∼1 in 4 substantial civil legal problems [LSC 2022, verify] | 0.30 | 0.20–0.40 | ∼30% become aware of recourse | Civil legal need; not algorithmic-decision awareness specifically |

| Barrier (layer) | Source domain | Exact statistic [source] | $p$ | Range | Mapping rule | Transportability limitation |
|---|---|---|---|---|---|---|
| Record Access Barriers (L3) | Credit reporting | Statutory free-report access with practical barriers [FCRA §612; CFPB 2022, verify] | 0.55 | 0.45–0.65 | Most navigable barrier (∼55%) | FCRA access; not clinical record access |
| Legal Knowledge Gap (L3) | Civil legal | 39% believe they can use the legal system to protect their rights [LSC 2022, verify] | 0.25 | 0.15–0.35 | ∼25% know specific algorithmic rights | General legal confidence; not algorithm-specific rights knowledge |
| Legal Resource Barriers (L3) | Civil legal | 46% cite cost; ∼50% of applicants turned away [LSC 2022, verify] | 0.40 | 0.30–0.50 | ∼40% with knowledge access resources | Civil legal aid capacity; not algorithm-specific representation |
| Systemic Bias in Algorithms (L3) | Healthcare audit | Cost-for-need proxy bias; correction raises Black enrollment 17.7%→46.5% [Obermeyer 2019] | 0.30 | 0.20–0.40 | ∼30% recourse not defeated by residual system bias | Single commercial tool / one health system; NOT a population-wide bias rate (prior “62%” figure was erroneous and is corrected) |

## S2 Table. Individual barrier removal effects

**Table 3. Individual barrier removal effects.** Baseline $P = 0.0018\%$; $\Delta$ = marginal effect of removing each barrier (deterministic; $\Delta_j = P(1/p_j - 1)$, Proposition 1). All $\Delta < 0.02\%$; the maximum is Legal Knowledge Gap ($\Delta = 0.0054\%$). Bootstrap 95% CIs ($n = 1{,}000$, seed 42) are provided in the repository (`individual_barrier_effects.csv`).

| **Barrier** | **Layer** | $P$ **after removal (%)** | $\Delta$ **(%)** |
|---|---|---|---|
| Rapid Data Transmission | Data Integration | 0.0060 | 0.0042 |
| Multi-System Integration | Data Integration | 0.0033 | 0.0015 |
| Permanent Storage | Data Integration | 0.0040 | 0.0022 |
| Error Detection Difficulty | Data Accuracy | 0.0051 | 0.0033 |
| Correction Process Barriers | Data Accuracy | 0.0051 | 0.0033 |
| Incomplete Correction Propagation | Data Accuracy | 0.0045 | 0.0027 |
| Awareness Gap | Institutional | 0.0060 | 0.0042 |
| Record Access Barriers | Institutional | 0.0033 | 0.0015 |
| Legal Knowledge Gap | Institutional | 0.0072 | 0.0054 |
| Legal Resource Barriers | Institutional | 0.0045 | 0.0027 |
| Systemic Bias in Algorithms | Institutional | 0.0060 | 0.0042 |

## S3 Table. Layer interaction decomposition (corrected)

**Table 4. Baseline-relative factorial decomposition of achievable improvement.** Values match the code, Fig 3 heatmap, and S-figures. (Corrects a stale main-text table in the prior submission.)

| **Effect** | **Share of achievable improvement (%)** |
|---|---|
| L1 (Data Integration) | 0.02 |
| L2 (Data Accuracy) | 0.03 |
| L3 (Institutional) | 0.36 |
| L1 $\times$ L2 | 0.44 |
| L1 $\times$ L3 | 4.51 |
| L2 $\times$ L3 | 7.03 |
| L1 $\times$ L2 $\times$ L3 (three-way) | 87.60 |
| Total | 100.0 |

**Table 5. Shapley value attribution (seeded, seed 42).** Fair allocation of achievable improvement across barriers, accounting for all removal orderings. Contributions are comparatively even (no barrier >11%) and span all three layers, consistent with interaction-driven structure.

| Rank | Barrier | Layer | Shapley (%) |
|---|---|---|---|
| 1 | Legal Knowledge Gap | Institutional | 10.95 |
| 2 | Rapid Data Transmission | Data Integration | 10.72 |
| 3 | Systemic Bias in Algorithms | Institutional | 10.53 |
| 4 | Incomplete Correction Propagation | Data Accuracy | 10.10 |
| 5 | Awareness Gap | Institutional | 9.67 |
| 6 | Error Detection Difficulty | Data Accuracy | 9.53 |
| 7 | Correction Process Barriers | Data Accuracy | 9.21 |
| 8 | Legal Resource Barriers | Institutional | 8.50 |
| 9 | Permanent Storage | Data Integration | 7.79 |
| 10 | Record Access Barriers | Institutional | 6.79 |
| 11 | Multi-System Integration | Data Integration | 6.21 |

**Table 6. Sobol first-order ($S_1$) and total-order ($S_T$) indices (seeded, seed 42).** $S_T$ exceeded $S_1$ for *all* barriers, but the $S_T - S_1$ interaction gaps are small and, for several barriers, within Monte-Carlo error at the current Saltelli sample size—so "substantially exceeded for all barriers" is not supported. Under seeding the smallest gaps fall on Correction Process/Legal Resource, underscoring that per-barrier orderings are sampling-sensitive (finalization option: increase $n$).

| Barrier | $S_1$ | $S_T$ | $S_T - S_1$ |
|---|---|---|---|
| Rapid Data Transmission | 0.063 | 0.101 | 0.038 |
| Multi-System Integration | 0.073 | 0.103 | 0.030 |
| Permanent Storage | 0.059 | 0.097 | 0.038 |
| Error Detection Difficulty | 0.069 | 0.098 | 0.029 |
| Correction Process Barriers | 0.088 | 0.103 | 0.015 |
| Incomplete Correction Propagation | 0.074 | 0.101 | 0.027 |
| Awareness Gap | 0.075 | 0.100 | 0.025 |
| Record Access Barriers | 0.075 | 0.108 | 0.033 |
| Legal Knowledge Gap | 0.084 | 0.103 | 0.019 |
| Legal Resource Barriers | 0.094 | 0.106 | 0.011 |
| Systemic Bias in Algorithms | 0.065 | 0.098 | 0.034 |

**Table 7. Stability of key findings within the prespecified ±10% perturbation scenario** ($n = 1{,}000$ resamples). Reported as stability within bounded scenarios, not robustness under arbitrary uncertainty; see S7 for coefficient-of-variation escalation.

| Finding | Threshold | % resamples supporting |
|---|---|---|
| Three-way share high | >70% | 100.0 |
| Max individual effect small | <1% | 100.0 |
| Barriers needed for 90% success | ≥10 of 11 | 100.0 |

## S4 Table. Shapley value attribution

## S5 Table. Sobol sensitivity indices

## S6 Table. Robustness under bounded perturbation

## S7 Table. Signal-to-noise and output variability

## S-Copula Table. Robustness to alternative topologies (Major 4)



At $\rho = 0.5$ the best single-layer intervention (4.50%) remains far below coordinated removal (96.8% total), so the comparative/threshold conclusion holds across the

**Table 8. SNR and coefficient of variation (CV) by parameter-noise level (seeded, seed 42).** Robustness is bounded: CV rises steeply above ∼15% noise, reaching 109.8% at 25% and 128.7% at 30%. Robustness is therefore claimed only within bounded scenarios.

| Noise (%) | SNR (dB) | Quality | CV (%) |
|---|---|---|---|
| 1 | 14.8 | Good | 3.3 |
| 2 | 11.7 | Good | 6.8 |
| 5 | 7.8 | Marginal | 16.6 |
| 10 | 4.6 | Marginal | 35.1 |
| 15 | 2.8 | Marginal | 53.0 |
| 20 | 1.4 | Marginal | 73.0 |
| 25 | −0.4 | Poor | 109.8 |
| 30 | −1.1 | Poor | 128.7 |

Seeded canonical values (seed 42). These replace the prior unseeded values; the output is small and right-skewed, so CV point estimates are sampling-sensitive at the current $n$ (finalization option: increase $n$ to stabilize).

**Table 9. Correlated-barrier (Gaussian copula) and repeated-attempt robustness** (seed 42, $n = 100{,}000$). The comparative conclusion (single-layer ≪ coordinated) is robust to correlation; the baseline value is correlation-sensitive; repeated attempts attenuate interaction dominance, delimiting the domain of validity.

| Scenario | Baseline $P$ | 3-way share | max 1-barrier | max 1-layer |
|---|---|---|---|---|
| Independence ($\rho = 0$) | 0.0018% | 87.6% | 0.0054% | 0.36% |
| Copula $\rho = 0.1$ | 0.049% | 84.5% | 0.071% | 0.97% |
| Copula $\rho = 0.2$ | 0.288% | 81.9% | 0.185% | 1.85% |
| Copula $\rho = 0.3$ | 0.835% | 79.9% | 0.376% | 2.74% |
| Copula $\rho = 0.4$ | 1.778% | 78.5% | 0.606% | 3.71% |
| Copula $\rho = 0.5$ | 3.200% | 77.4% | 0.821% | 4.50% |
| Block copula $\rho = 0.5$ | 0.151% | 66.0% | 0.142% | — |
| Repeated attempts $m = 2$ | 0.352% | 53.3% | 0.452% | 5.70% |
| Repeated attempts $m = 3$ | 3.651% | 25.2% | 2.665% | 16.88% |

correlation sweep; under repeated attempts it attenuates, bounding the model's validity to approximately single-pass serial recourse.

## Supplementary figures

## Code and data availability

All code, data, and a one-command `make` reproduction pipeline (seed 42) are at `https://github.com/Nyx-Dynamics/algorithmic-bias-epidemiology-academic`, archived at Zenodo (DOI:10.5281/zenodo.18746745). Key scripts: `barrier_visualization.py`, `sensitivity_analysis.py`, `copula_robustness.py`, `build_derivation_table.py`. Environment: Python 3.10+, NumPy, SciPy, Matplotlib. Random seed 42 (now enforced in all stochastic routines).

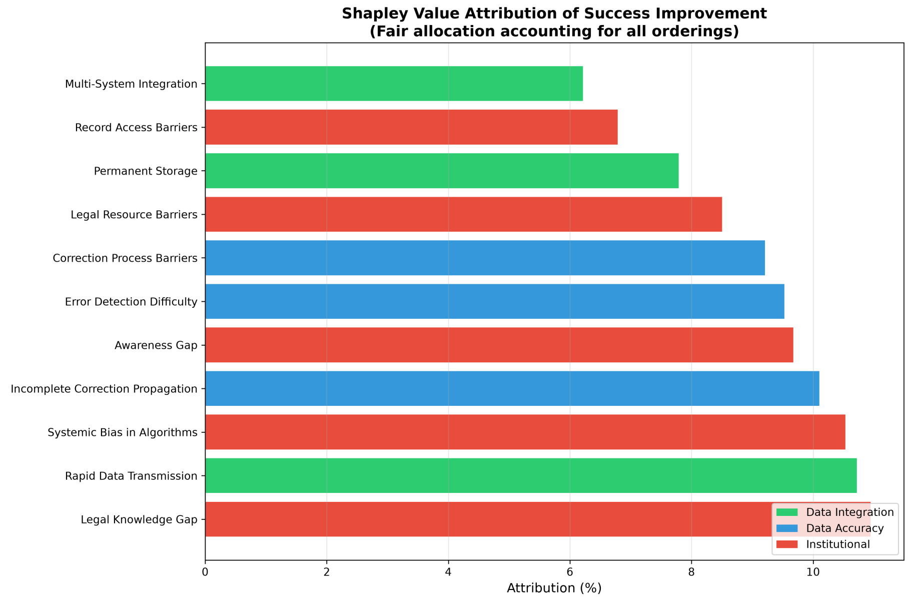


**Figure 1. S1 Fig. Shapley value attribution (seeded).** Fair allocation of achievable improvement across barriers, accounting for all removal orderings; the five highest contributors span all three layers, consistent with interaction-driven structure.

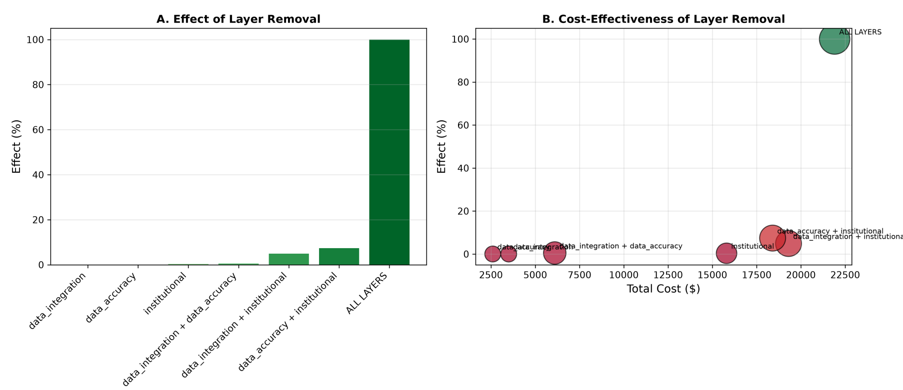


**Figure 2. S2 Fig. Layer-removal effects.** Single-layer removal yields negligible improvement; the largest two-layer combination is Data Accuracy + Institutional (∼7.4%); only removal across all three layers reaches 100%.

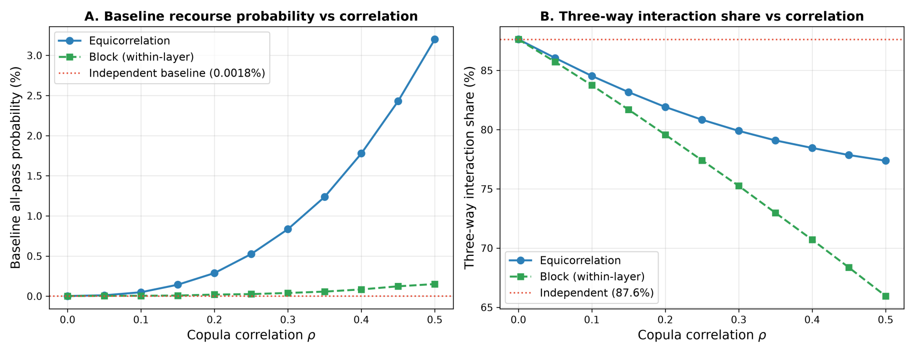


**Figure 3. S3 Fig. Alternative-topology robustness (Major 4).** Baseline success $P$ and three-way interaction share vs. Gaussian-copula correlation $\rho$, with repeated-attempt points ($m = 2, 3$). The comparative conclusion is robust to correlation; the baseline value is correlation-sensitive; repeated attempts attenuate interaction dominance.